

# React for Kids

## Intro to Components

**Subject:** Web Development with React

**Level:** Beginner - Kids / Teens

**Duration:** 45-60 minutes

**Context:** India-themed analogies (tiffin dabba, mithai, train)

**Topics:** Components, JSX, App Component, Reusability

---

### Learning Objectives:

- Understand what React is and why it was created
- Define a component and explain it using real-life Indian analogies
- Identify the three parts of a React component (function, return, JSX)
- Write a simple functional component with a return statement
- Reuse components inside an App component
- Appreciate how reusability saves time in coding

# Table of Contents

1. What is React? | Pg 3
2. What is a Component? | Pg 3
3. Anatomy of a React Component | Pg 4
4. Code Example: Greeting Component | Pg 4
5. Reusing Components | Pg 5
6. The App Component | Pg 5
7. Code Example: App + Greeting | Pg 6
8. Guided Practice | Pg 6
9. Mini-Challenge | Pg 7
10. Recap & Key Terms | Pg 7
11. Bonus: Fun React Facts! | Pg 8
12. Exit Ticket | Pg 8

# 1. What is React?

React is a free, open-source JavaScript library created by engineers at Facebook in 2013. It helps developers build fast, interactive web applications by breaking the user interface into small, manageable pieces called components. Today, React powers apps like Instagram, Netflix, and WhatsApp Web.

**Lego Analogy:** Think of React like a box of Lego bricks. Each brick is a component. You snap them together to build something amazing and you can reuse the same bricks in different projects!

## Key Facts:

- **Created by:** Facebook (Meta) engineers - released publicly in 2013
- **Type:** JavaScript library (not a full framework)
- **Specialty:** Building reusable UI components
- **Used by:** Instagram, Netflix, Airbnb, WhatsApp Web, and thousands more

# 2. What is a Component?

A component is a small, self-contained piece of a webpage. Each component handles one job showing a heading, a button, an image, or a product card. When you put many components together, you get a complete web application.

**Tiffin Dabba:** Each compartment of a tiffin box holds one food item—dal, rice, sabzi, roti. Each React component holds one piece of the page.

**Train Compartments:** An Indian train has compartments for sleeping, sitting, and luggage. Each compartment has its own job, yet together they form one train.

**Rubber Stamp:** A rubber stamp lets you print the same design over and over. Components let you write code once and reuse it everywhere.

### 3. Anatomy of a React Component

Every functional React component has three essential parts. Understanding each part will help you read and write components confidently.

1. **Function Declaration:** Defines the component as a JavaScript function. The name **MUST** start with a Capital Letter—just like your name in the class attendance register!  
`function Greeting() {`
2. **Return Statement:** The return keyword sends back the HTML-like content that will appear on the screen. Think of the sweet-wala handing you a gulab jamun—the return statement hands the page its content.  
`return <h1>Namaste!</h1>;`
3. **JSX (JavaScript XML):** JSX looks like HTML but lives inside JavaScript. React converts it into real DOM elements. You can write tags inside JS!

## 4. Code Example: Greeting Component

Let's look at our very first complete React component. Read each line carefully and notice how the three parts we just learned appear in the real code.

```
// A simple React component
function Greeting() {
  return (
    <h1>Namaste! Welcome to React!</h1>
  );
}
```

## 5. Reusing Components - The Rubber Stamp Magic

One of React's superpowers is reusability. Once you write a component, you can use it as many times as you want—just like stamping the same ink design on multiple pages without redrawing it every time.

```
// Using Greeting three times
function App() {
  return (
    <div>
      <Greeting /> { /* stamp 1 */}
      <Greeting /> { /* stamp 2 */}
      <Greeting /> { /* stamp 3 */}
    </div>
  );
}
```

## Why Reusability Matters:

Imagine you have 50 product cards on a shopping website. Without components you'd write the same HTML 50 times. With React, write it ONCE — use it 50 times. Change one component → every instance updates automatically!

## 6. The App Component (Your Mithai Box)

Every React project has a special component called App. It is the top-level container—the mithai box that holds all the other components. When React starts your app, it looks for App and renders it first.

## 7. Code Example: App Using Greeting Components

```
// 1. Define the Greeting component
function Greeting() {
  return <p>Namaste! Welcome to React!</p>;
}

// 2. Define the App component (the mithai box)
function App() {
  return (
    <div>
      <h1>My First React App</h1>
      <Greeting /> {/* use Greeting here */}
      <Greeting /> {/* use it again! */}
    </div>
  );
}
```

**Output on screen:** My First React App | Namaste! Welcome to React! x2

## 8. Guided Practice

Complete the code below to write your first React component. Remember: the function name must start with a Capital Letter!

```
function _____() {  
  return <h2>Namaste!</h2>;  
}
```

**Q1. What should go in the blank? (Hint: it should start with a capital letter)**

Answer: Greeting (or any Capitalized name)

**Q2. What keyword sends content back to the screen?**

Answer: return

**Q3. What do we call the HTML-like code inside JavaScript?**

Answer: JSX

**Q4. If you wanted to show this greeting twice, how would you write it in App?**

Answer: <Greeting /> twice inside the container div.

## 9. Mini-Challenge: Build Your Namaste Card

Now it's your turn to be a React developer! Follow the two steps below to build your very own Namaste Card app.

**Step 1 - Create a NameCard Component:** Write a NameCard component that displays your name inside a `<p>` tag. Use function `NameCard()` and return your name.

```
function NameCard() {  
  return <p>My name is Codey</p>;  
}
```

**Step 2 - Combine Components in App:** Put both the Greeting component AND your new NameCard component inside the App component.

```
function App() {  
  return (  
    <div>  
      <Greeting />  
      <NameCard />  
    </div>  
  );  
}
```

**Bonus Challenge:** Can you add a third component called FavSubject that shows your favourite school subject? Place it inside App after NameCard!



## 10. Recap & Key Terms Glossary

- **React:** A JavaScript library for building fast, component-based UIs
- **Component:** A small, reusable piece of a webpage (function + return + JSX)
- **JSX:** HTML-like syntax used inside JavaScript in React
- **Reusability:** Write a component once; use it as many times as needed
- **App:** The root component that holds all other components inside it
- **Capital Name:** React component function names MUST start with a capital letter
- **Return:** The keyword that sends content to be displayed on the page

## 11. Bonus: Fun React Facts!

- **Born in 2013:** React was created by Jordan Walke at Facebook and open-sourced in May 2013.
- **Instagram:** Instagram's entire web app was rewritten in React—one of the earliest big adoptions.
- **WhatsApp Web:** The web version of WhatsApp uses React for its fast, real-time message updates.
- **Virtual DOM:** React uses a 'Virtual DOM'—an invisible copy of the page to figure out the smallest possible update needed.
- **Netflix:** Netflix uses React on its TV UI—fast rendering is critical for a smooth watching experience.
- **Popularity:** React is consistently ranked as the most-used front-end library by developers worldwide.

## 12. Exit Ticket

Before you leave today's lesson, answer these questions in your own words.

**Easy:** What is a React component? Use one of today's analogies in your answer.

---

**Medium:** What are the three parts of a React component? Give a one-line description of each.

---

**Hard:** Why is reusability important in programming? Give a real-life example.

---

**Star:** Can a component go inside another component? How did we do this today?

---

**Great work, young coder! Keep exploring React!**



## React for Kids: Extra Practice Worksheet

---

*For Grade 4 Students — Test Your React Superpowers! (No Answers)*

**Question 1:** Which famous social media company created React in 2013?

.....

**Question 2:** Name two huge applications you or your parents might use that are built using React.

.....

**Question 3:** Fill in the missing word: An Indian train has separate compartments for sleeping, sitting, and luggage. In web design, we break a page into similar small parts called \_\_\_\_\_.

.....

**Question 4:** What is the strict rule about naming a component function? (Hint: Think about writing your name in a school register!)

.....

**Question 5:** What keyword must you use inside a component function to send content back onto the computer screen?

.....

**Question 6:** What does "JSX" look like when we write it inside our JavaScript code?

---

**Question 7:** If you use a component like a "Rubber Stamp" to show a button 50 times on a page, what happens if you later change the color inside that single component file?

---

**Question 8:** Look closely at this code snippet. What is the spelling mistake that will break the component?

```
Function Laddu() {  
  return <h1>Sweet!</h1>;  
}
```

---

**Question 9:** Why is the special App component called the "Mithai Box" of our project?

---

**Question 10:** How many times will "Crunchy Samosa!" show up on the page based on the App component below?

```
function App() {  
  return (  
    <div>  
      <Samosa />  
      <Samosa />  
    </div>  
  );  
}
```

**Question 11:** Write the code tag you would use to insert a component named FavSubject inside the App component.

**Question 12:** React uses an invisible copy of the website called the Virtual \_\_\_\_\_ to make pages load super-fast!